PHYS 7501:
Particles and Nuclei I

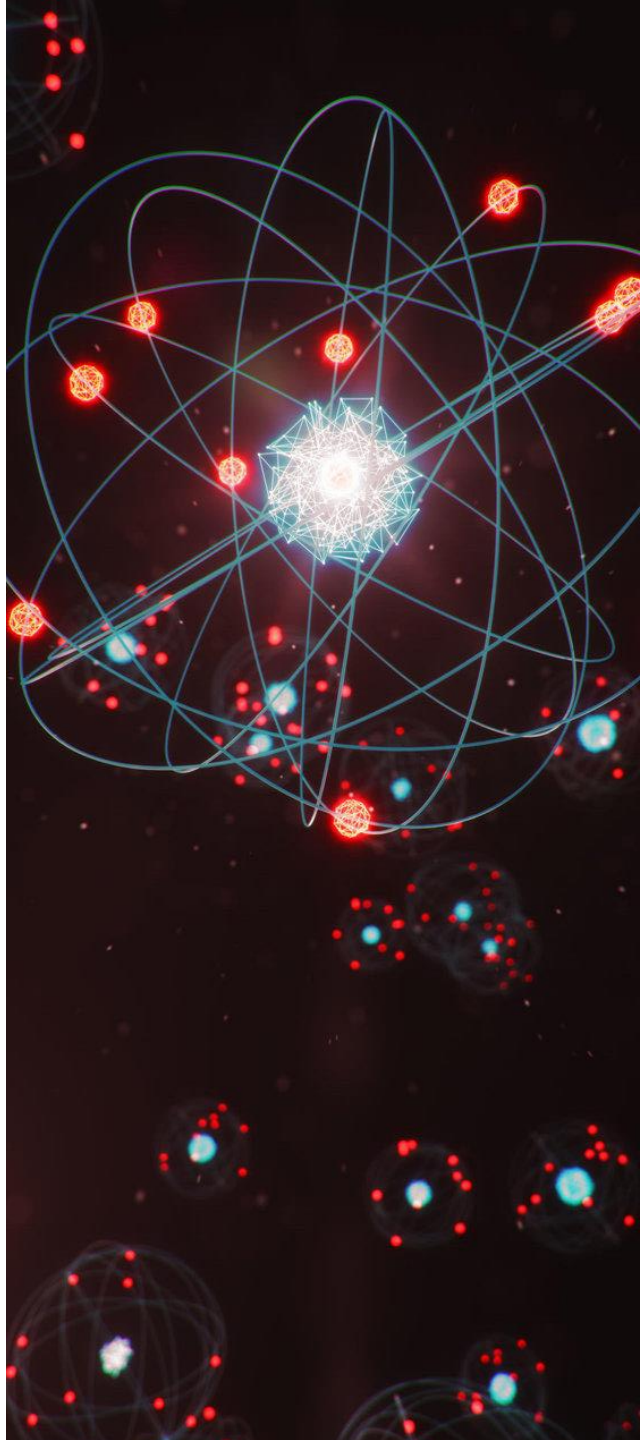
Tues./Thurs. 2 – 3:20PM
Clippinger 105

Dr. Andrea Richard

Edwards Accelerator Lab 204

richarda1@ohio.edu

Available via e-mail & Teams

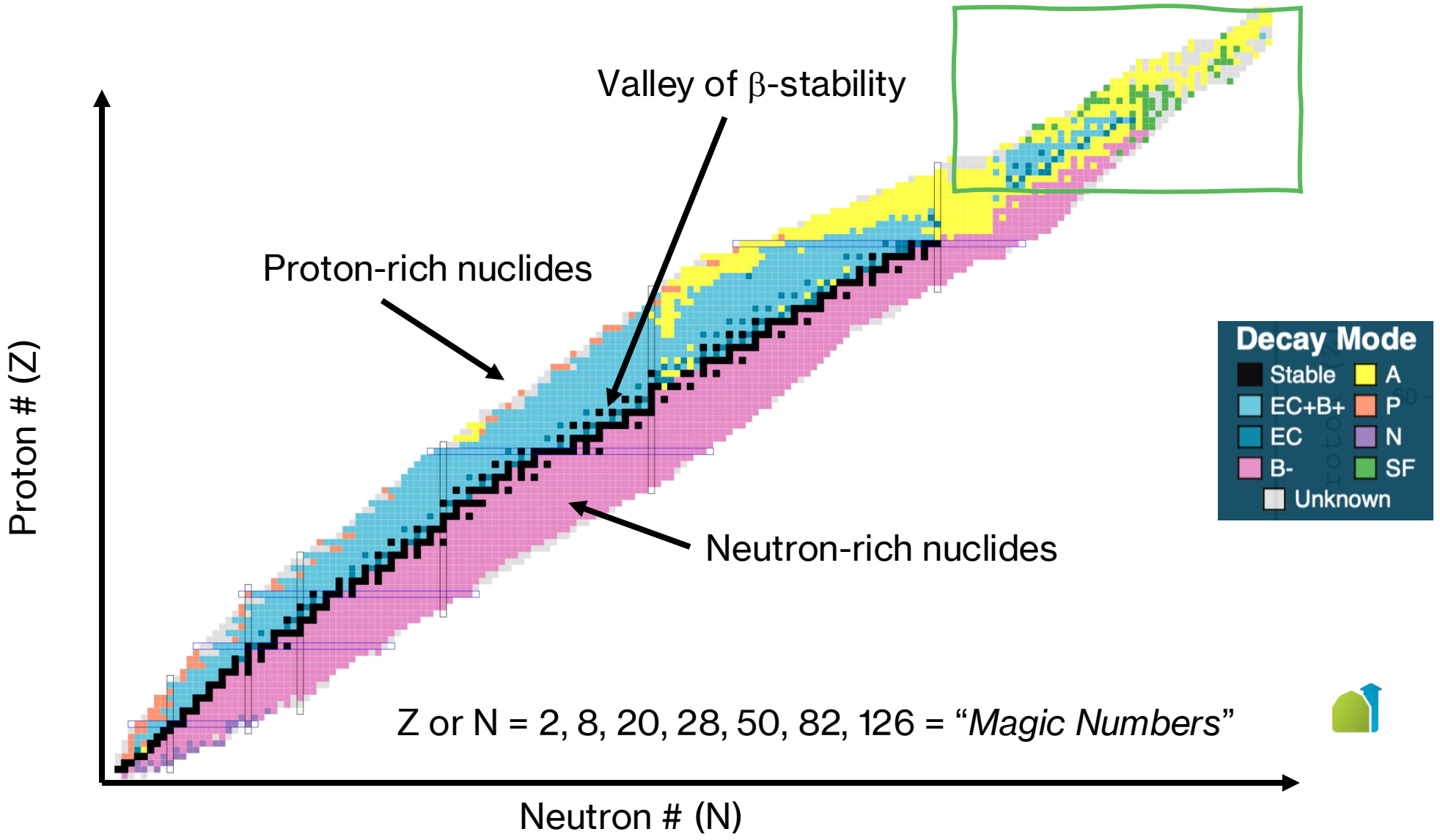


FISSION

Modern Nuclear Chemistry, Chapter 11

Monday	Tuesday	Wednesday	Thursday	Friday
10/6	10/7	10/8	10/9	10/10
	Exam Day		**Fission**	*Fall Break*
10/13	10/14	10/15	10/16	10/17
	Nuclear Reactions HW#3 Due		**Scattering**	
10/20	10/21	10/22	10/23	10/24
	Direct Reactions		**Resonant Reactions**	

Radioactive Decay in the Chart of the Nuclides

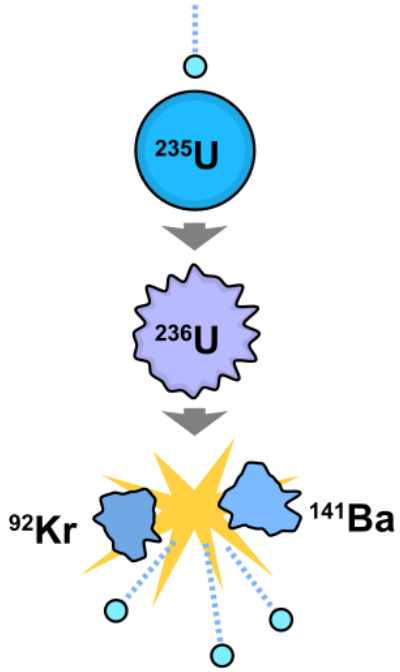


Historical Perspective

- In 1939, Otto Hahn and Fritz Strassmann (chemists, experimental) discovered that when uranium was irradiated with neutrons, alkaline earth metals were produced
- Lise Meitner and Otto Frisch her nephew (physicists, theoretical) then suggested that when uranium absorbs a neutron, it becomes sufficiently excited to split into two fragments.
- Neils Bohr and John Wheeler worked out the Liquid Drop Model theory to explain the mechanism of fission
- In 1942, Enrico Fermi and his team achieved the first controlled nuclear chain reaction: a milestone for nuclear power and the Manhattan Project



Lise Meitner and Otto Hahn

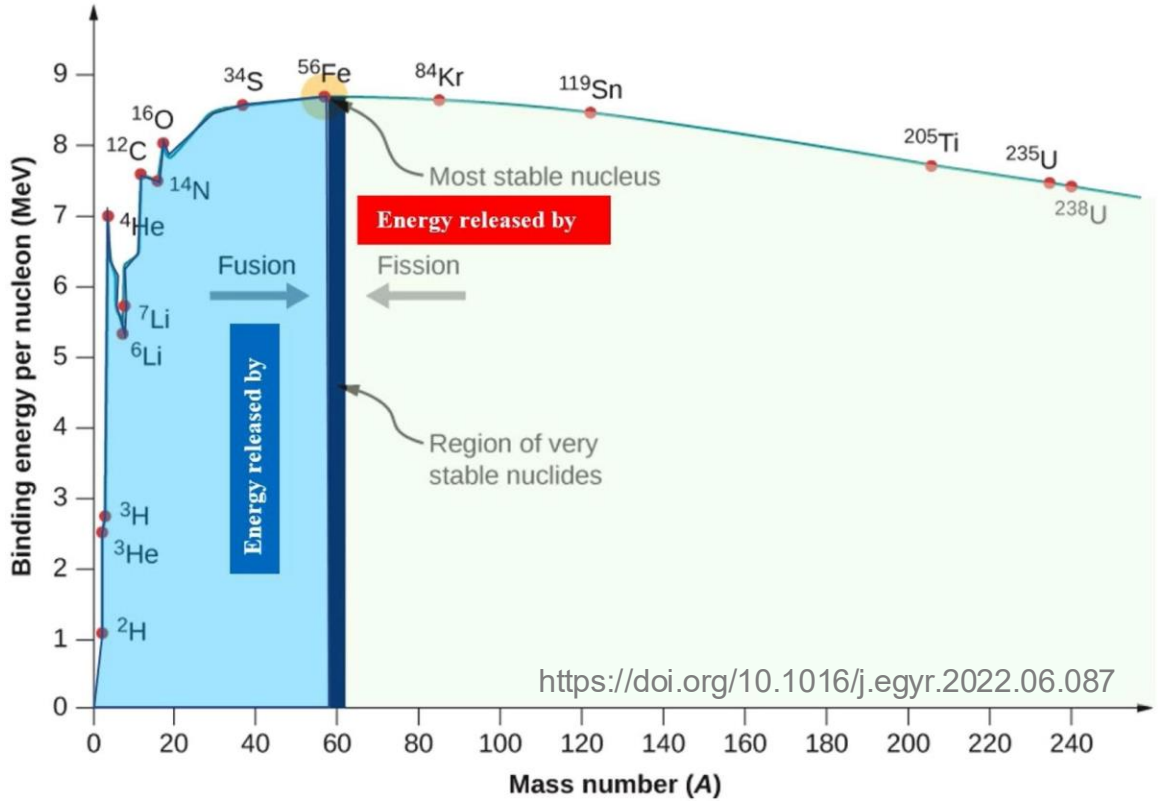


Binding Energy and Fission

- For $A \approx 120$, BE/A ~ 8.5 MeV/u
- In the Uranium region, BE/A ~ 7.6 MeV/u
- If a heavy particle split into two fragments, the increased binding energy per particle would be released as kinetic energy of the fragments given by other forms of radiation
- Simplistically, use $A \approx 240$, and assume the nucleus splits into two equal fragments with $A \approx 120$ then the total energy released would be

$$240(8.5 - 7.6) = 220\ \text{MeV}$$

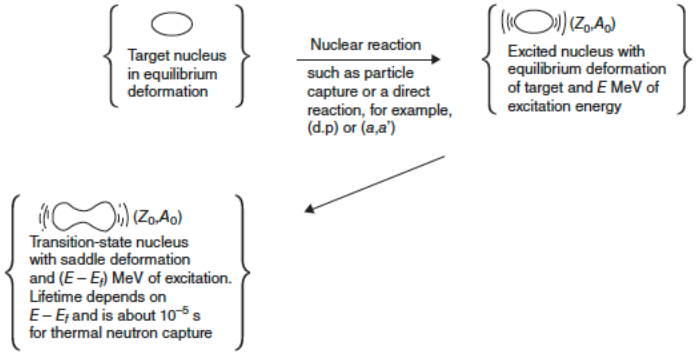
- The final state has ~220 MeV in energy that will be shared between the fragments, neutrons, γ-rays, β-particles, etc.



The Fission Process

A complicated process that no one theory can truly describe

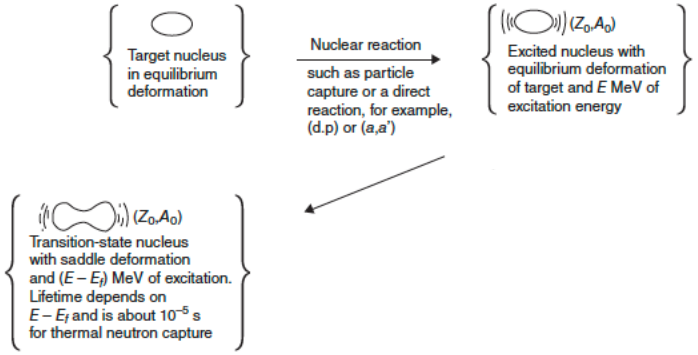
- Nucleus with some small ground-state deformation absorbs energy (via some process) and becomes excited. It deforms into a “transition state” or “saddle point” configuration



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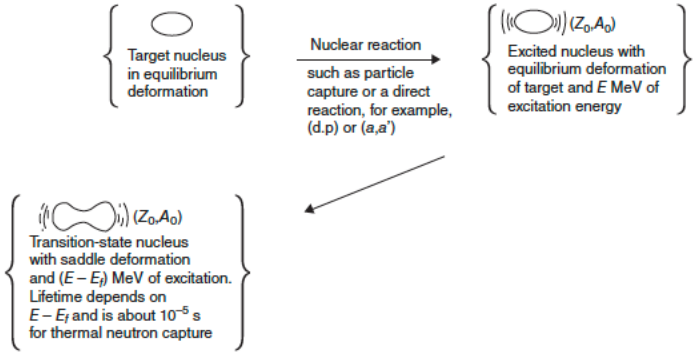
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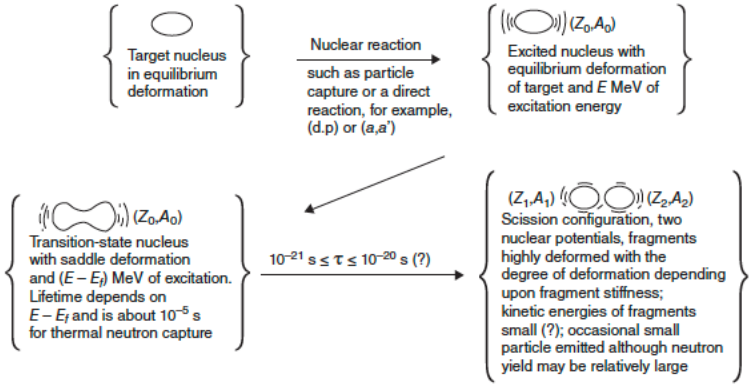
- Nucleus with some small ground-state deformation absorbs energy (via some process) and becomes excited. It deforms into a “transition state” or “saddle point” configuration
- As the nucleus deforms, the Coulomb energy decreases (distance between protons increases), while nuclear surface energy increases
- At the saddle point, the rate of change in Coulomb is equal to the rate of change in the surface
- The formation and decay of the transition state of the nucleus is the rate-determining step in the fission process and corresponds to the passage over an activation energy barrier to the reaction



The Fission Process

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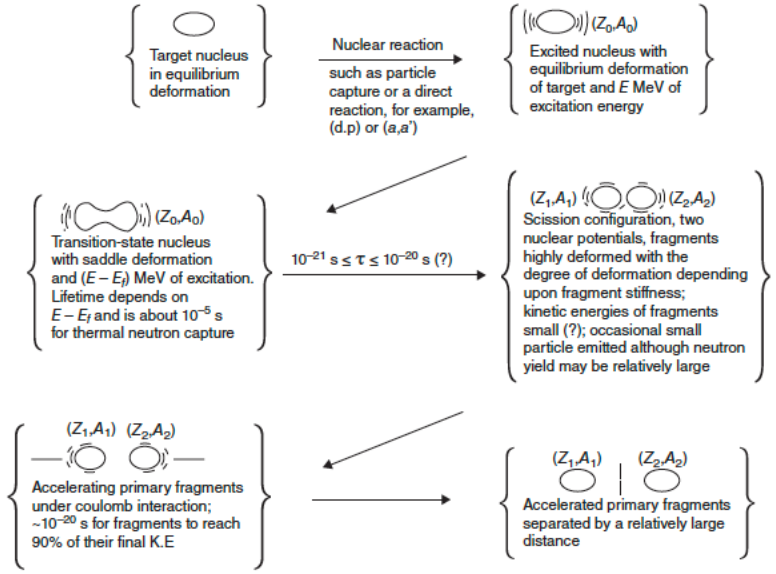
- When the nucleus crosses the transition state, then in a very short time, the neck between the nascent fragments disappears (ruptures) and the nucleus divides into two fragments at the “scission point.”
- At the “scission point” there are two deformed nuclear fragments in contact with each other



The Fission Process

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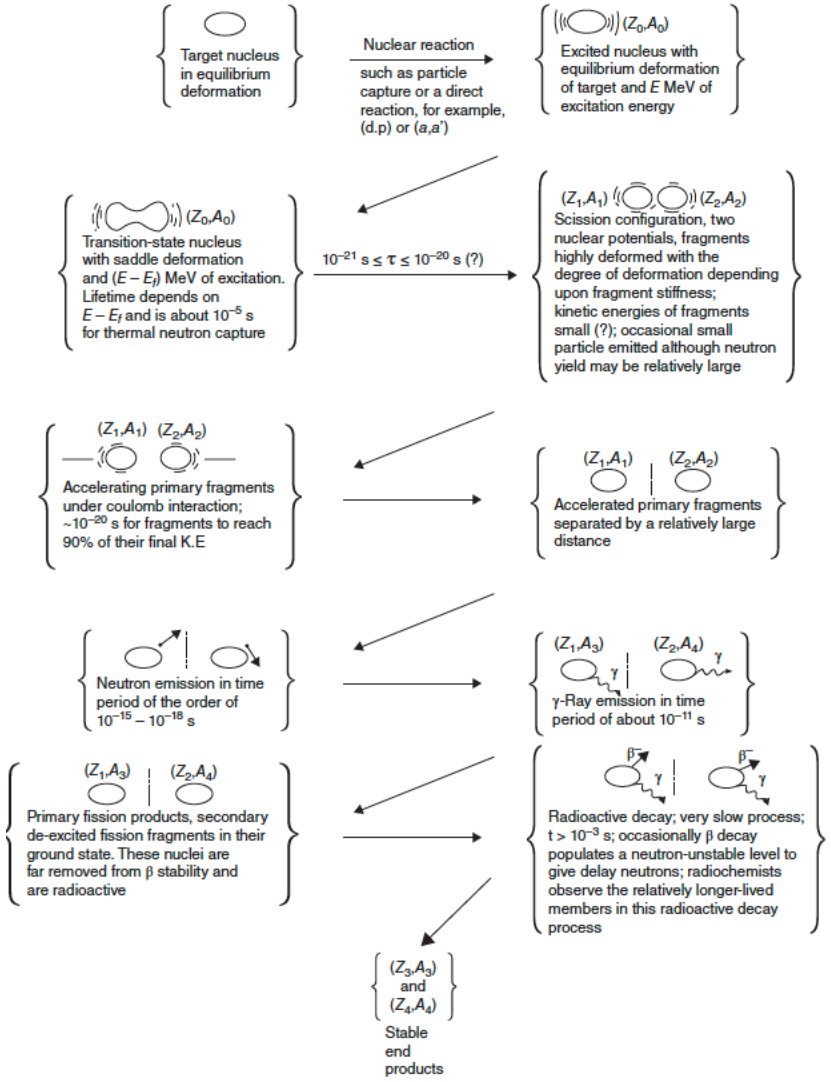
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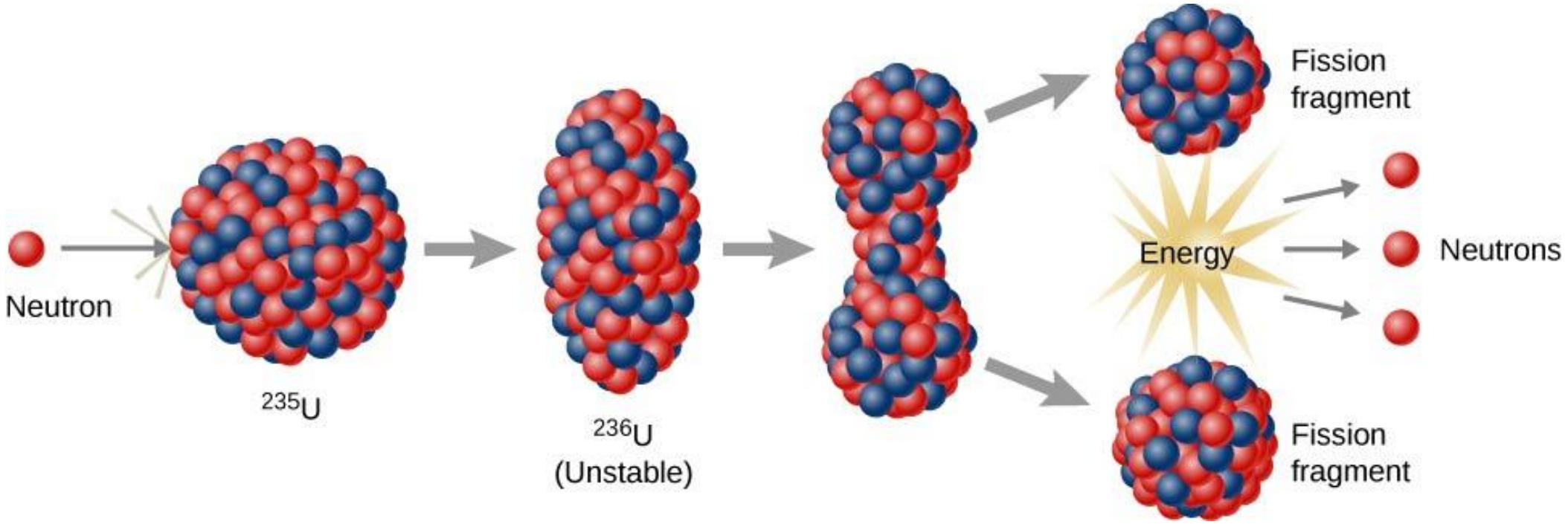
The Fission Process

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- At the “scission point” there are two deformed nuclear fragments in contact with each other
- As they move away from each other, they become more spherical
- Excitation energy removed via e-mission of prompt neutrons
- On longer timescales, β-decay occurs followed by delayed neutron emission.



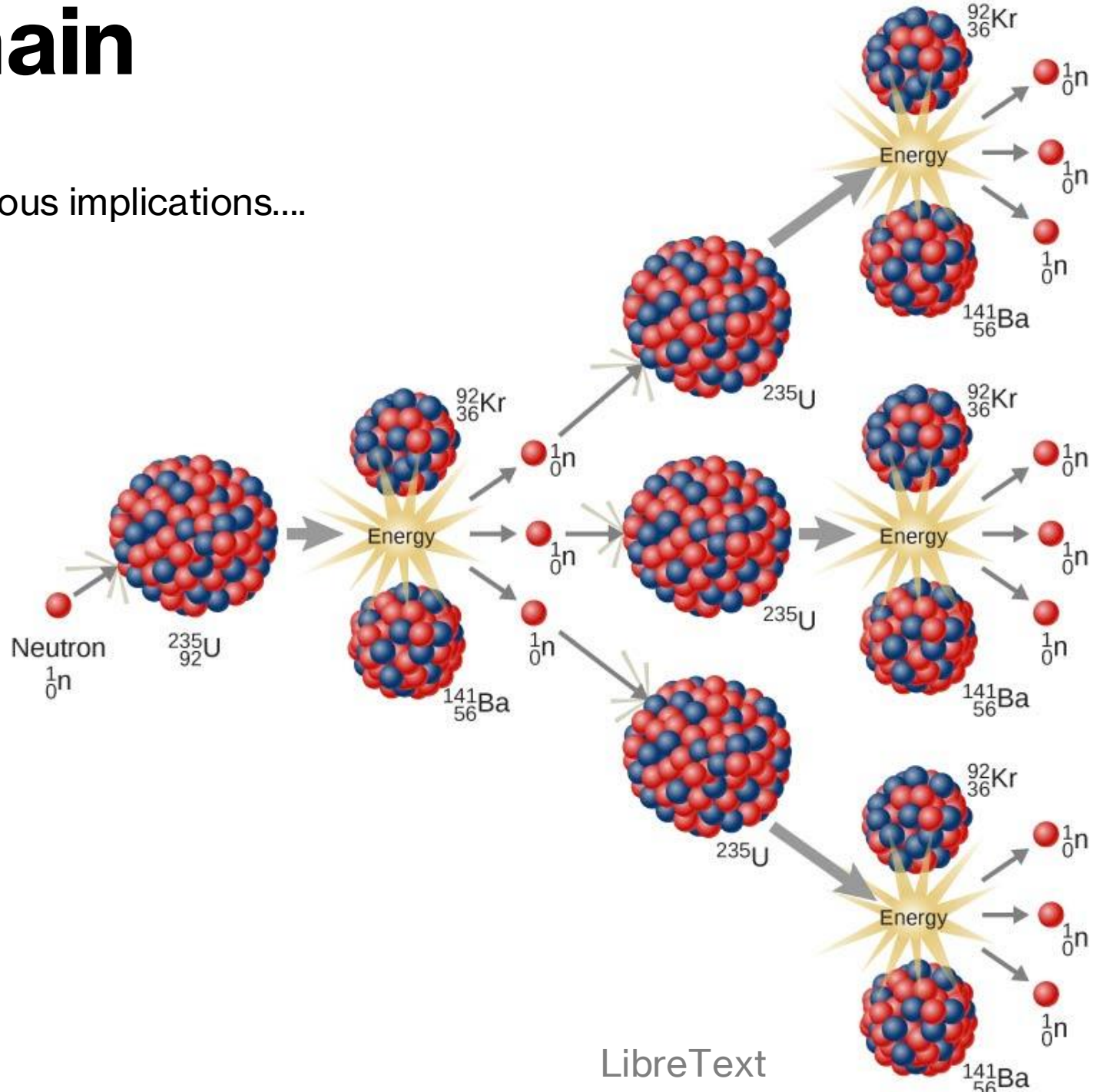
Cartoon Fission Process



LibreText

Fission Chain Reaction

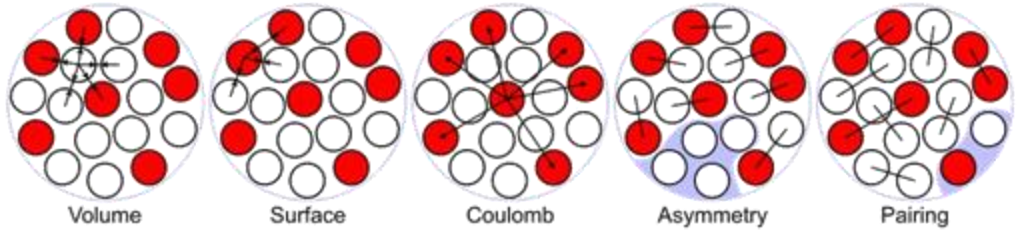
Some pretty serious implications....



LibreText

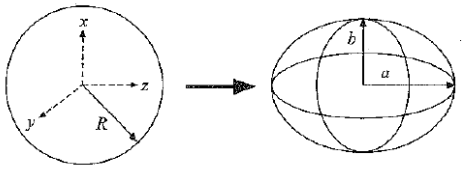
Probability of Fission: Liquid Drop Model

- Recall the contributions to the binding energy from the semi-empirical mass formula
- What happens if the nucleus deforms?
  - Volume is conserved, A-Z remains the same, pairing stays the same
  - But, the surface is enlarged and charges in the nucleus spread out



- Deformation will increase the binding energy penalty from the surface term, but decrease the binding energy penalty from the Coulomb term
- So to understand what happens to the overall binding energy when the nucleus is deformed, we need to understand how these two terms evolve under deformation
- The change in binding energy will be: $\Delta E = \mathrm{BE}_{\mathrm{final}} - \mathrm{BE}_{\mathrm{initial}} = \left(\mathrm{E}_{\mathrm{C}}^{'} + \mathrm{E}_{\mathrm{S}}^{'}\right) - (\mathrm{E}_{\mathrm{C}} - \mathrm{E}_{\mathrm{S}})$, where the prime terms include the modification to the Surface and Coulomb terms under deformation.

Probability of Fission: Liquid Drop Model



- The deformed shape (an ellipsoid) can be expressed using an expansion of the radius in terms of Legendre Polynomials

$$R = R_0[1 + \alpha_2 P_2 \cos\theta]$$ For axial symmetry, we only keep $\ell = 2$

α_2 is the quadrupole distortion parameter, which is related to the quadrupole deformation via $\alpha_2 = \sqrt{\frac{5}{4}\pi}\beta_2$

- The Coulomb and Surface terms can also be expanded in a similar way

$$E_C' \approx a_C \frac{Z^2}{A^{1/3}}\left(1 - \frac{1}{5}\alpha_2^2\right) \quad \text{and} \quad E_S' \approx a_S A^{2/3}\left(1 + \frac{2}{5}\alpha_2^2\right)$$

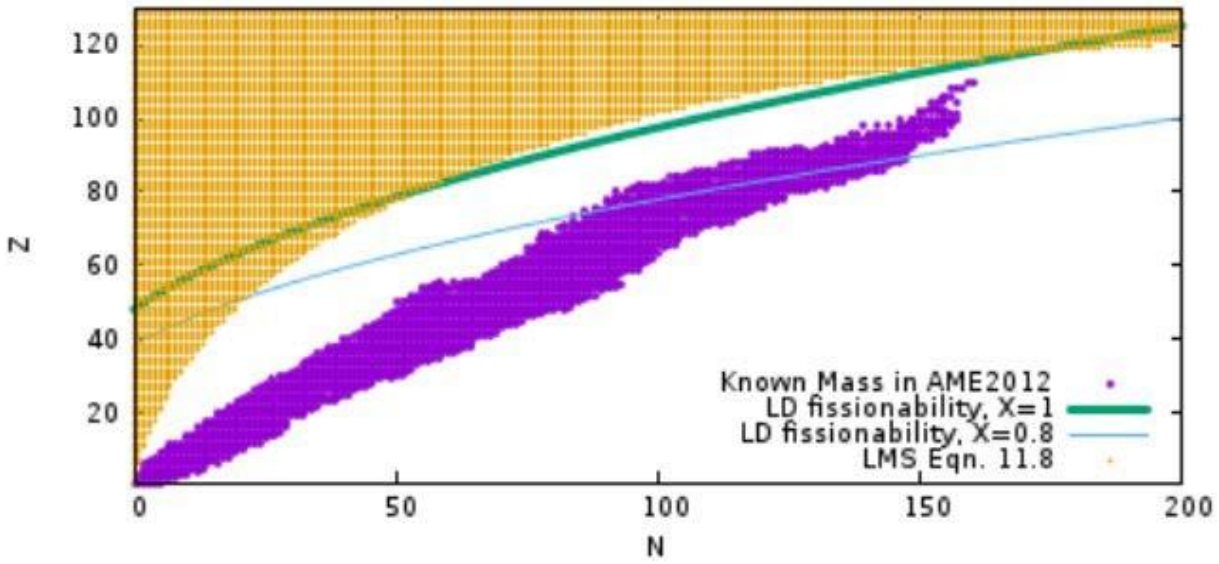
- Meaning that the cost for deformation is $\Delta E = \frac{\alpha_2^2}{5}\left(2a_S A^{2/3} - a_C \frac{Z^2}{A^{1/3}}\right)$ But, we already know this from the SEMF!

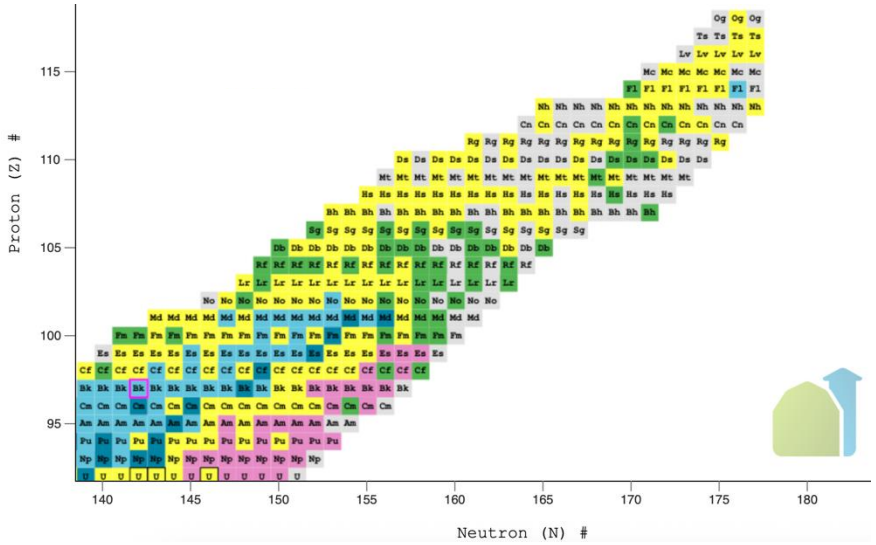
- So, when the non-deformed Coulomb energy is twice the non-deformed Surface energy, there is zero energetic cost to deform ⇒ Fission!

- The fissionability parameter is a measure of how favorable fission is: $x = \frac{E_C}{2E_s} = \frac{a_C}{2a_s}\frac{Z^2}{A} \equiv \frac{Z^2/A}{(Z^2/A)_{critical}}$

Probability of Fission: Liquid Drop Model

- What combination of Z, A is prone to fission? (i.e. What is $\left(Z^2/A\right)_{critical}$?)
  - For the LDM, we just defined $\left(\frac{a_C}{2a_s}\right)^{-1}$, which is $\approx$ **48** based on SEMF constants
    - This corresponds to $Z > 116, A \geq 270$
  - There is also a more sophisticated treatment: $\left(Z^2/A\right)_{critical} = 50.8333\left[1 - 1.7826\left(\frac{N-Z}{A}\right)^2\right]$
- Larger fissionability means a nucleus is more prone to fission





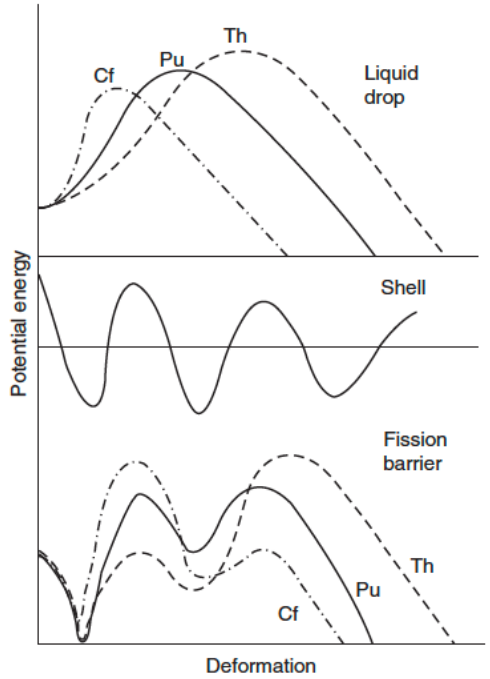
Fission Barrier

- To quantify the probability of fission occurring, we can consider the potential energy associated with a particular shape of the nucleus

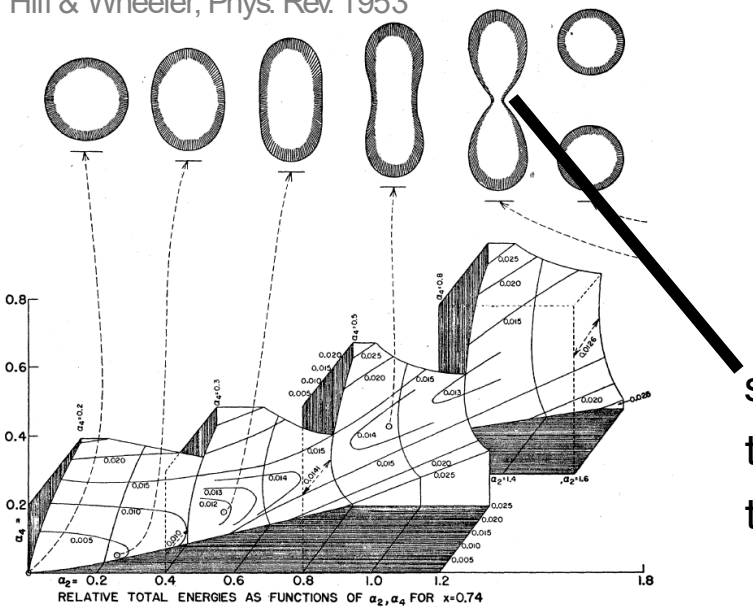
- This potential energy is dominated by liquid drop energetics,
  ...but with a major correction for enhanced binding near closed shells
  ...the magnitude of which is altered by the nuclear deformation

- So, the peaks and valleys in potential energy space are associated by the competition between liquid-drop and shell-model modified by deformation

- The likelihood of transitioning from one shape to another depends on the energetic cost associated with overcoming (or tunneling through) the barrier between them



Hill & Wheeler, Phys. Rev. 1953

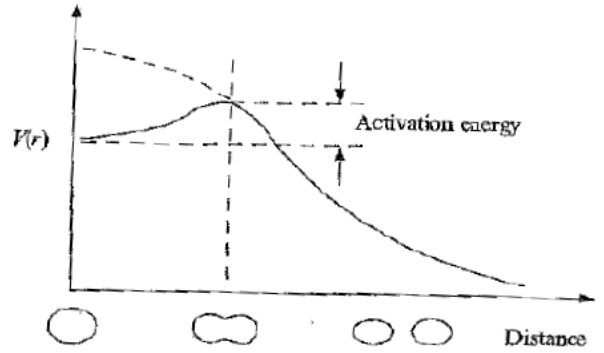


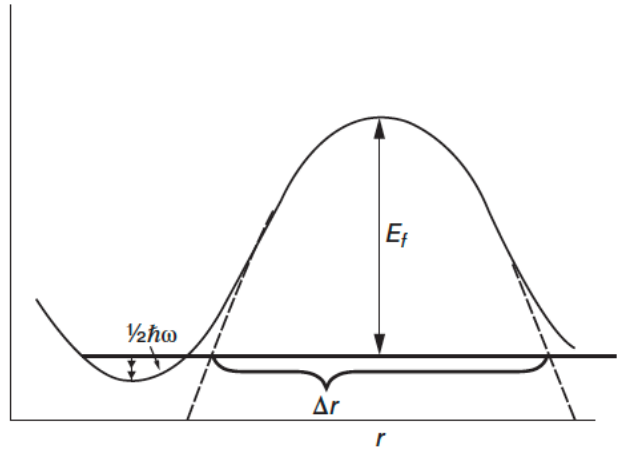
scission occurs at the saddle point in the potential

Fission Barriers & Activation Energy

- Quantitatively, the energy that must be ”paid” is an activation energy
- It’s more common to refer to a *fission barrier height* (E_f), which is measured relative to the ground state of the original system, quantified as a Harmonic Oscillator
- Rather than overcome the barrier, there is also a possibility to tunnel through it.
- The width of the barrier is related to $\hbar\omega$, where a large curvature means a thin barrier (easy to tunnel) and a small curvature means a thick barrier (harder to tunnel)
- Overcoming the barrier corresponds to *induced fission*, whereas tunneling through corresponds to *spontaneous fission*

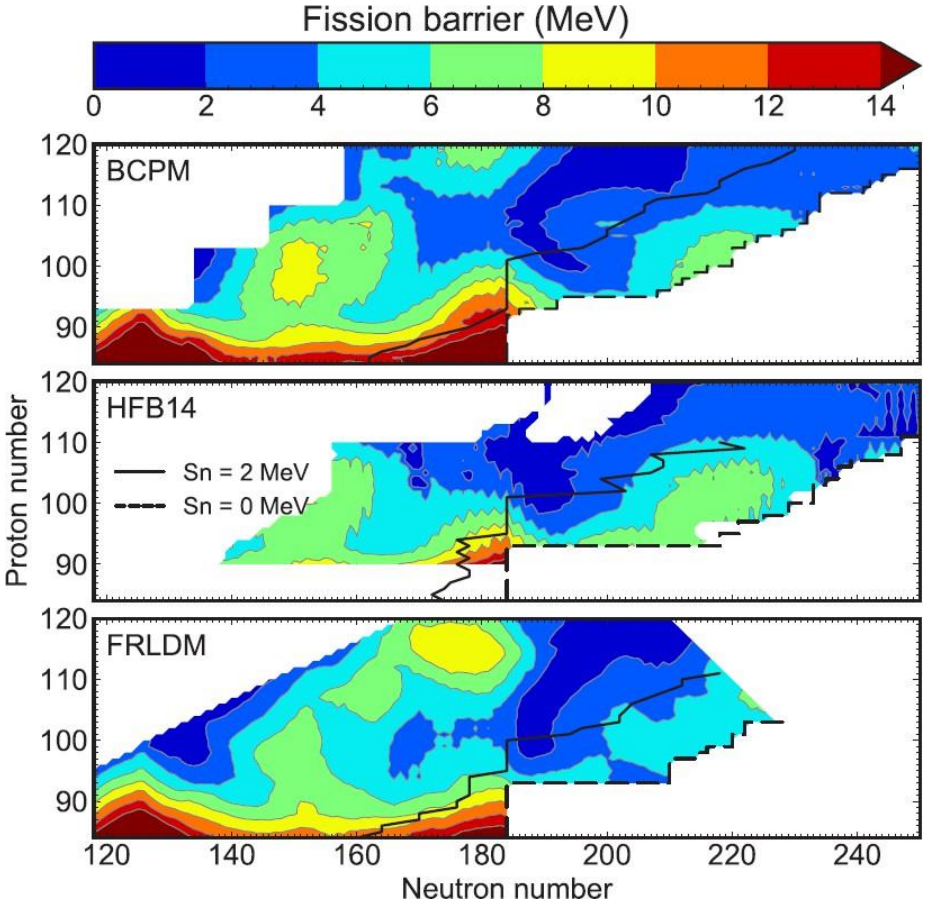
B.R. Martin, Nuclear and Particle Physics (2009)





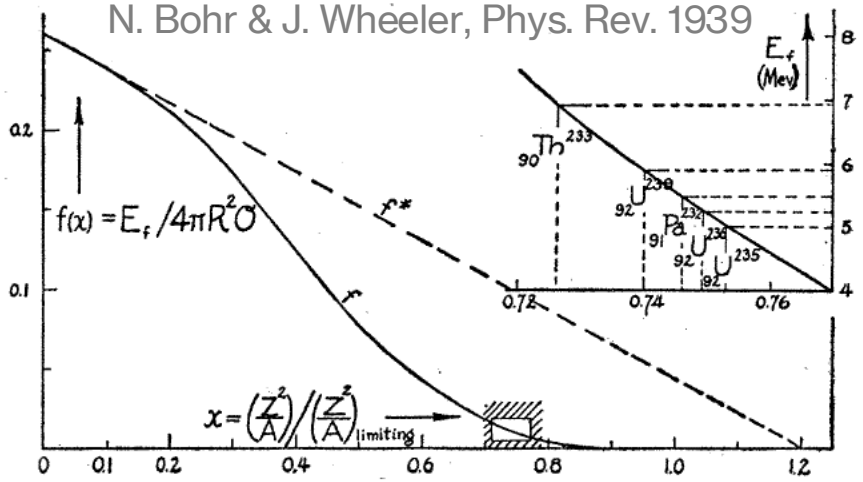
Fission Barrier Height

- In the LDM, the fission barrier height is the difference between the ground-state binding energy and the energy of two spherical nuclei made by splitting the system
- For uranium and thorium, this is usually around 5 – 7 MeV
- This is very hard to calculate and still an active area of research
- Requires sophisticated mass modelling beyond the scope of the class (e.g. FRDM)

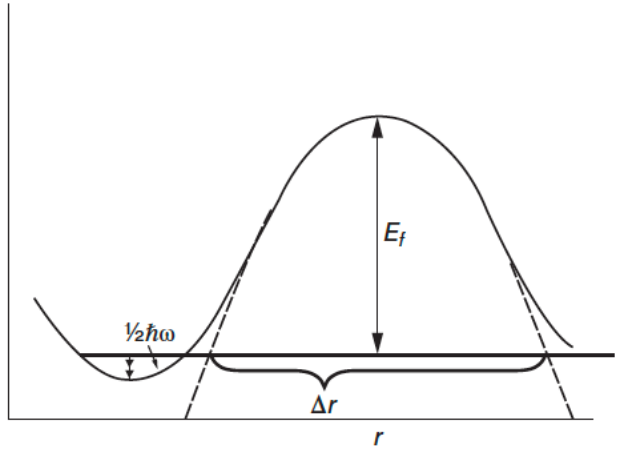


S. Giuliani et al. PRC 2018

N. Bohr & J. Wheeler, Phys. Rev. 1939



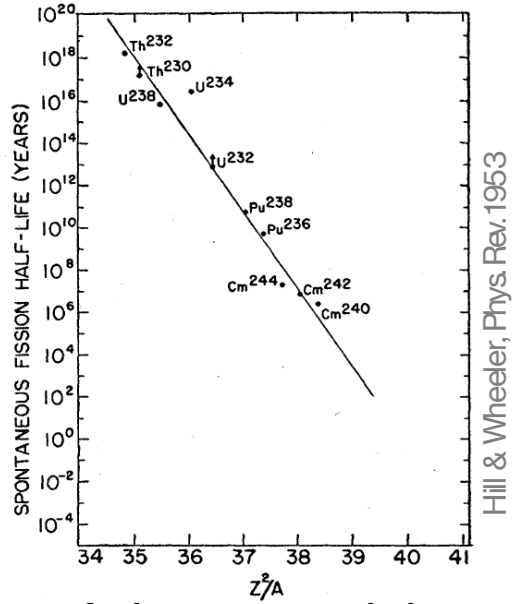
Spontaneous Fission Rate



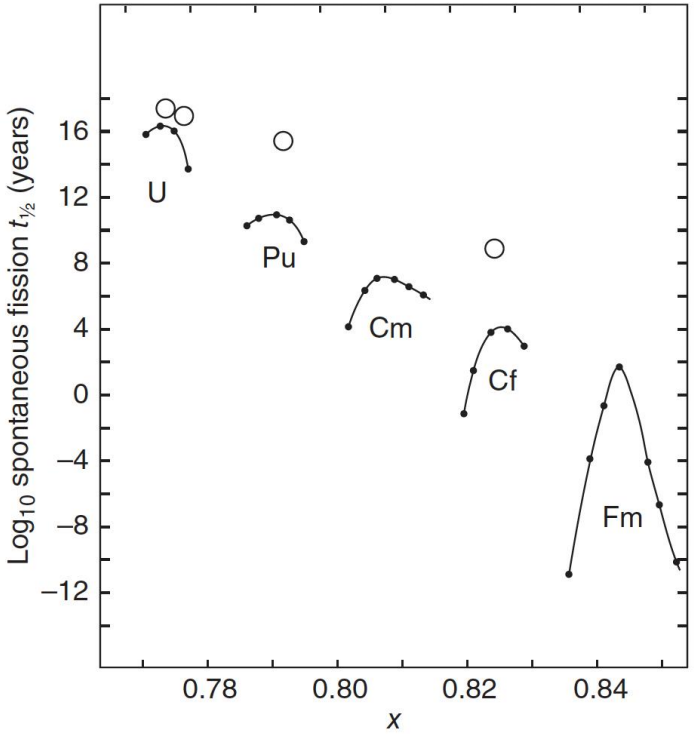
- Spontaneous Fission (SF) is similar to α decay, where a barrier is bombarded at some rate and there is a probability for tunneling through the barrier
- The difference now is that the potential is not from the nucleus but rather the energy surface for the landscape of possible shapes
- The nucleus tunnels through the barrier: $\boldsymbol{t_{1/2} = \frac{ln(2)}{fP}}$, where $\boldsymbol{f}$ is a bombarding frequency and $\boldsymbol{P}$ is the tunneling probability
- $\boldsymbol{f}$ corresponds to the rate at which the nuclear shape is changing
  - Surface oscillation frequency is $\sim 10^{20}/s$
- For the simplest case of a one-humped barrier, we approximate an inverted parabola with tunneling probability $\boldsymbol{P = \frac{1}{1+exp(2\pi E_f/\hbar\omega)}}$
- So, $\boldsymbol{t_{1/2} = 10^{-21} exp(2\pi E_f/\hbar\omega)}$

Spontaneous Fission Rate

- The height of the barrier is related to the fissionability parameter x
- So, we can actually determine the half-life for spontaneous fission empirically
- But this is very very rough

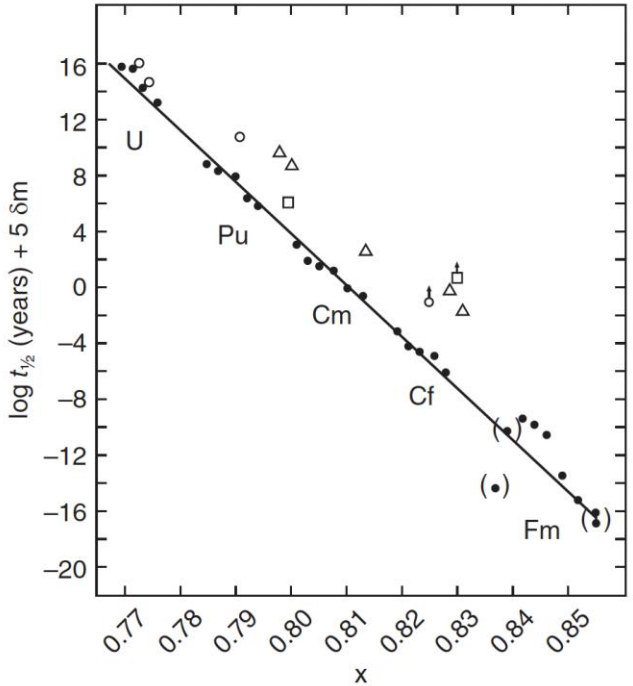


Hill & Wheeler, Phys. Rev. 1953

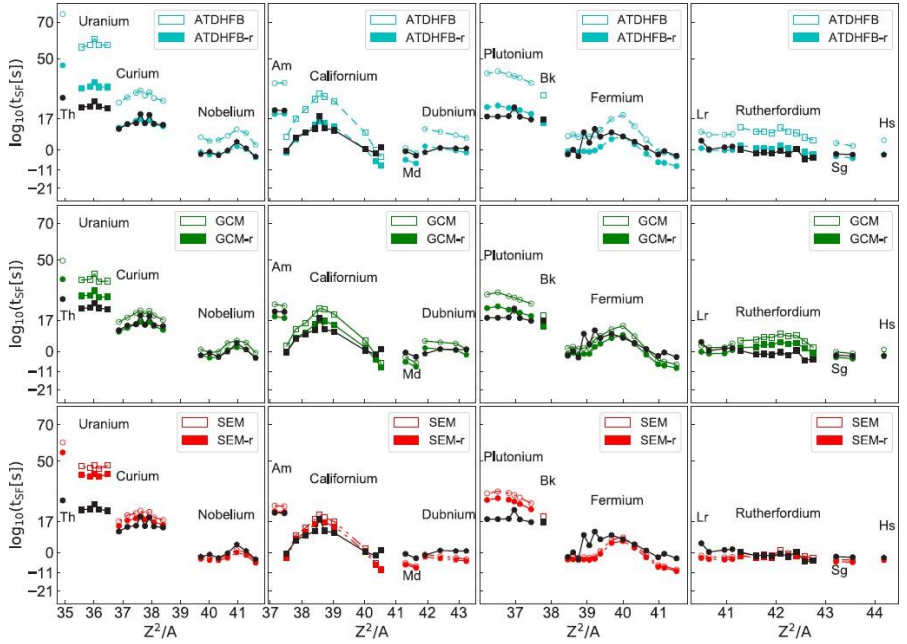


....using shell-corrected masses gives some improvement:

Vandenbosch & Huizenga, Nuclear Fission (1973)



Even state-of-the-art models struggle to accurately reproduce data:



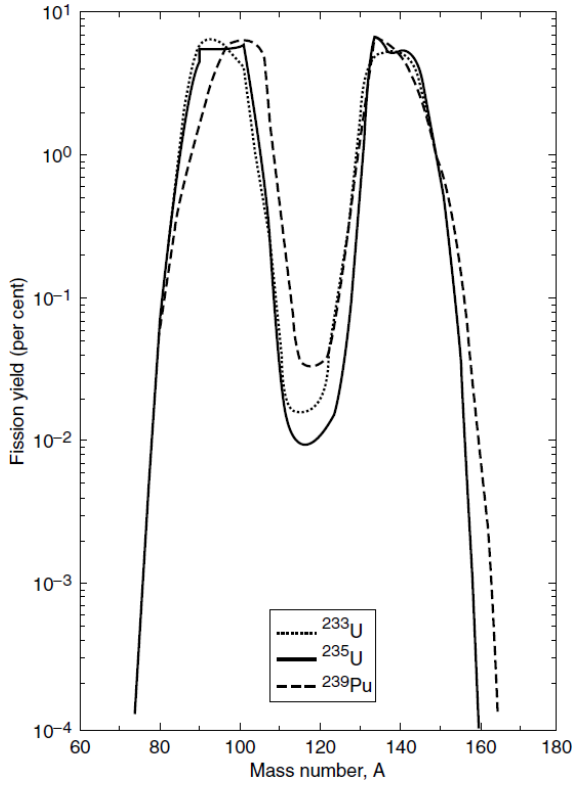
S. Giuliani et al. PRC 2018

Fission Fragment Mass Distributions

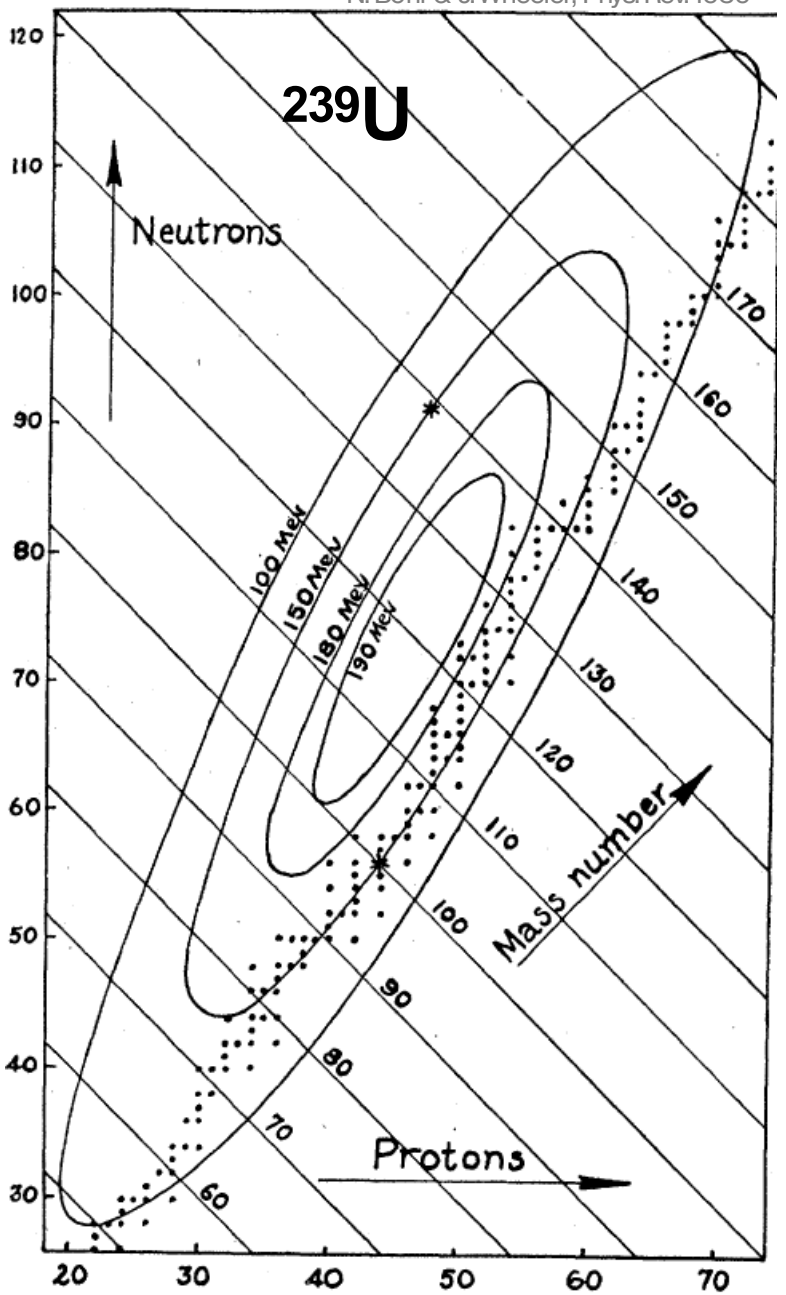
- To first order, the fission fragments will be those that maximize the total energy release (estimate from LDM)
- This would always result in a case of a symmetric distribution of the matter

- However, often the observed mass distributions are instead quite asymmetric:
- The individual fragments approximately maintain the original A/Z (because neutrons and protons separate in the same way), but often high-A and low-A peaks exist

Seaborg & Loveland, The Elements Beyond Uranium (1990)

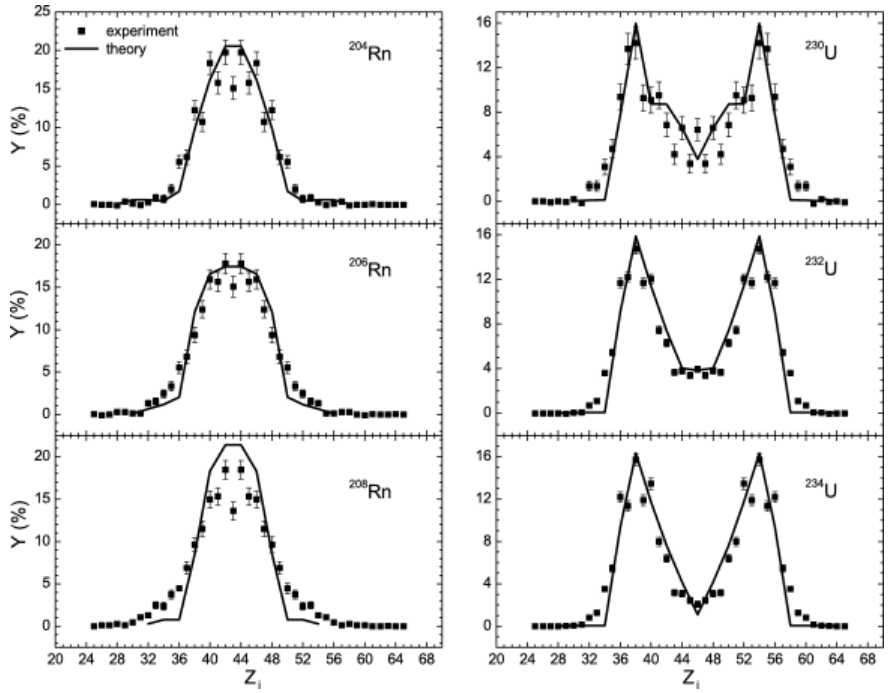


N. Bohr & J. Wheeler, Phys. Rev. 1939



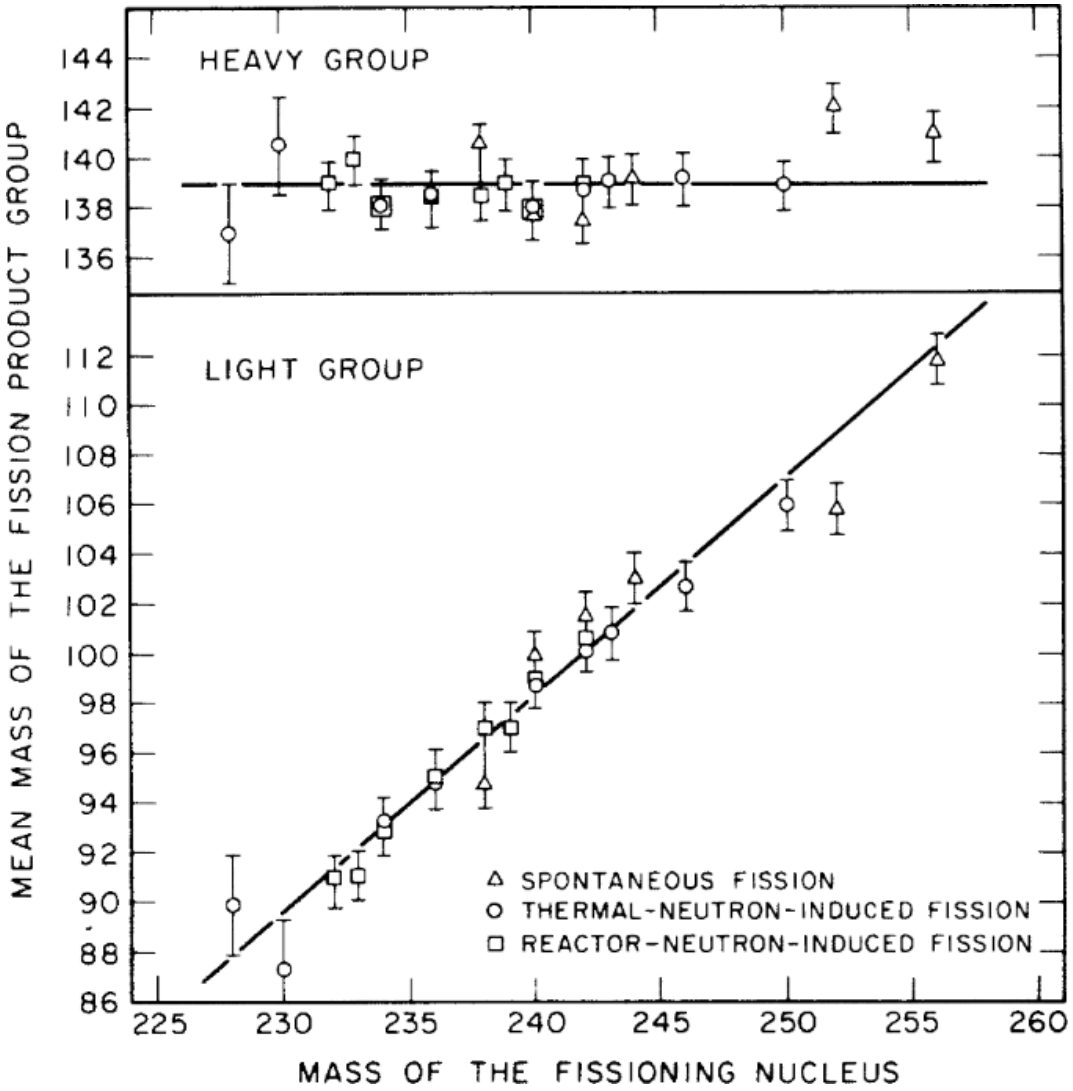
Fission Fragment Mass Distributions

- Why would the mass distribution be asymmetric?
- Magic numbers (of course!) at $Z = 50, N = 82$
- The other fragment has the remainder of *most* of the remaining nucleons



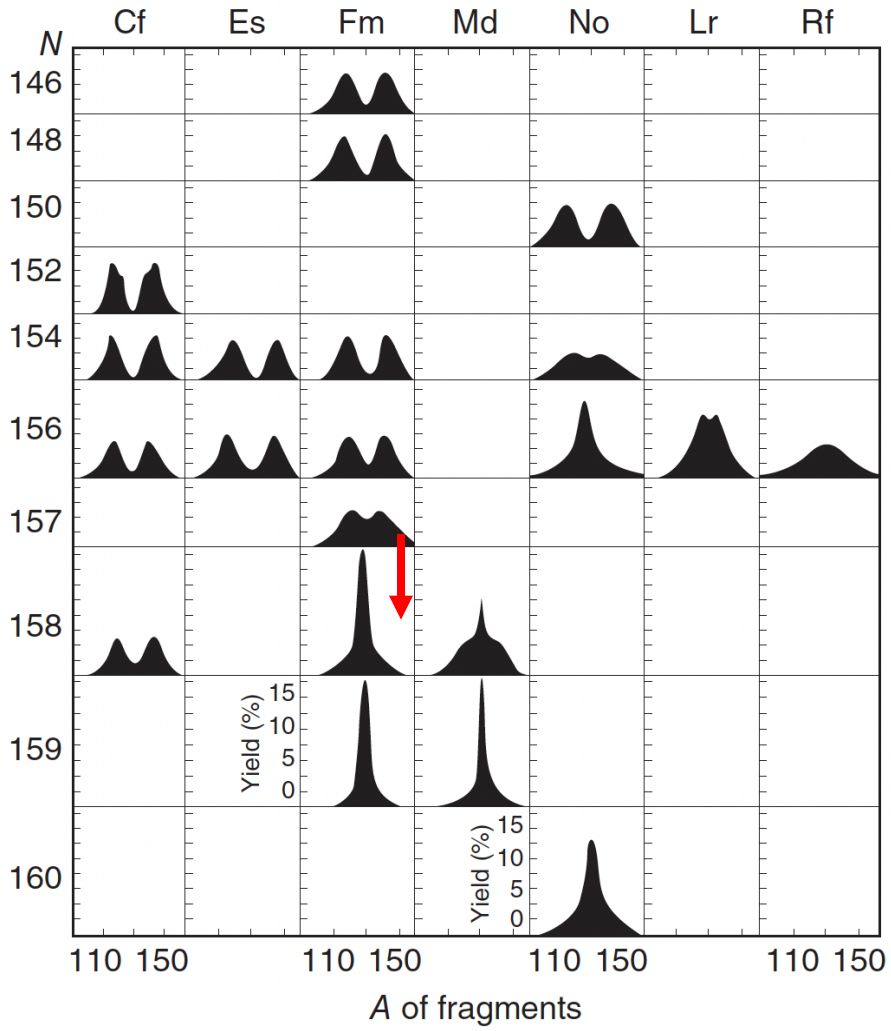
H. Pasca *et al*, PLB (2016)

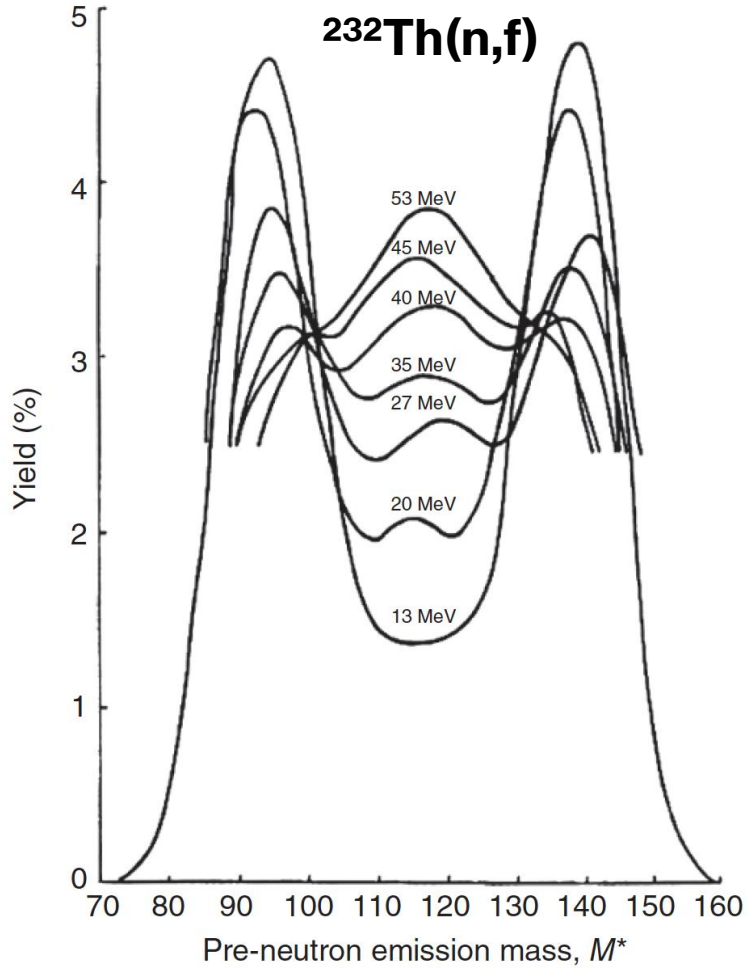
K. Flynn et al. PRC Vol 5 1972



Fission Fragment Mass Distributions

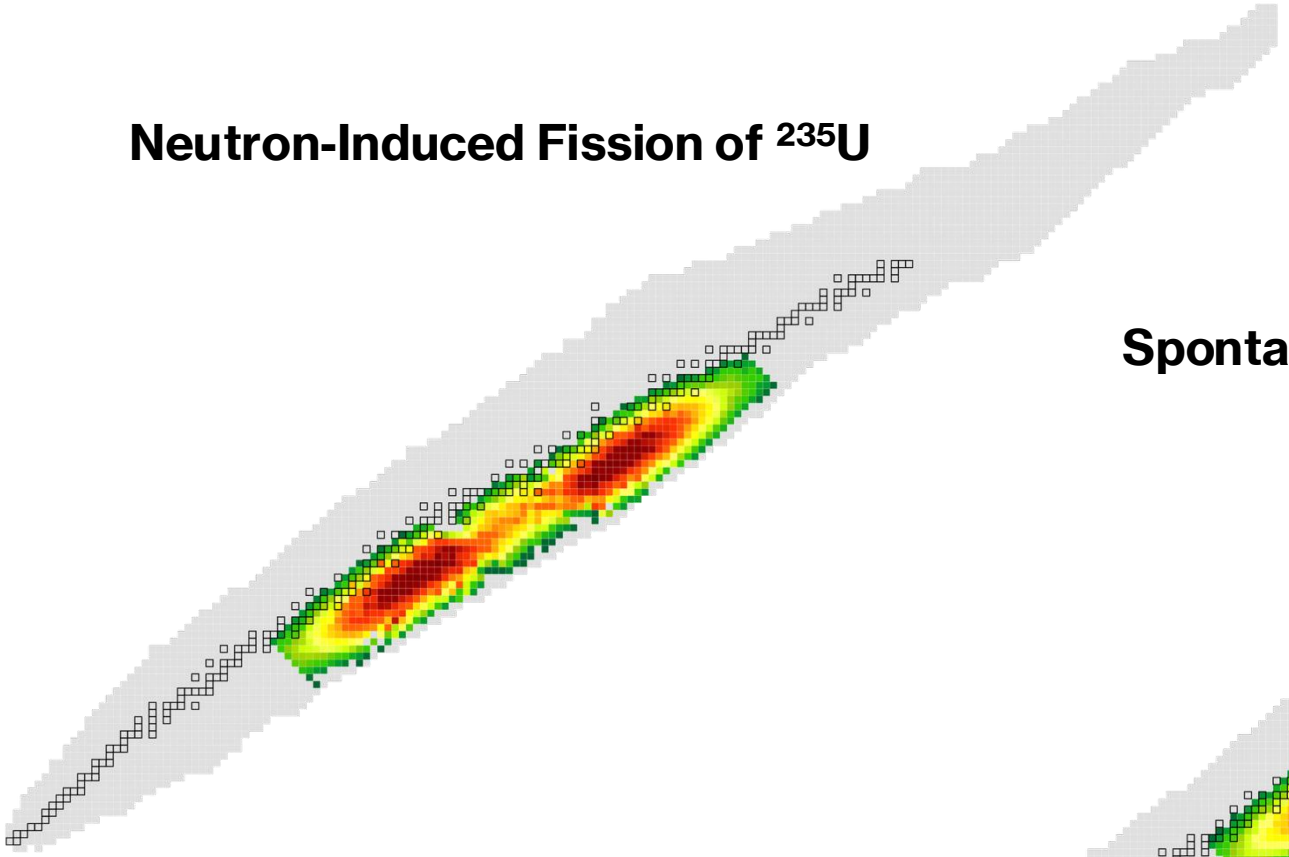
- However, the signature of closed shells can be erased if the symmetric fragment distribution is far from shell closures or if the excitation energy is too high



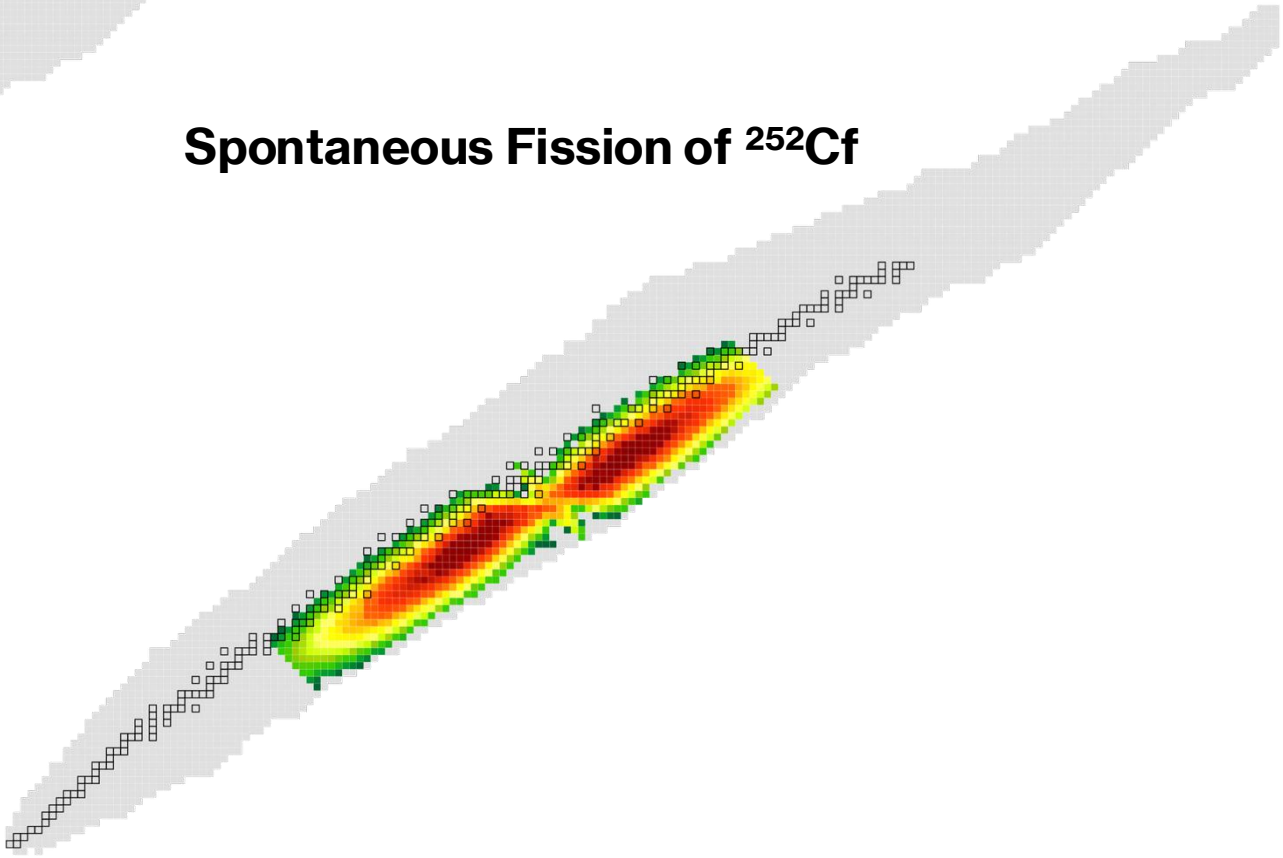


Example of Fission Fragment Distributions

Neutron-Induced Fission of ^{235}U

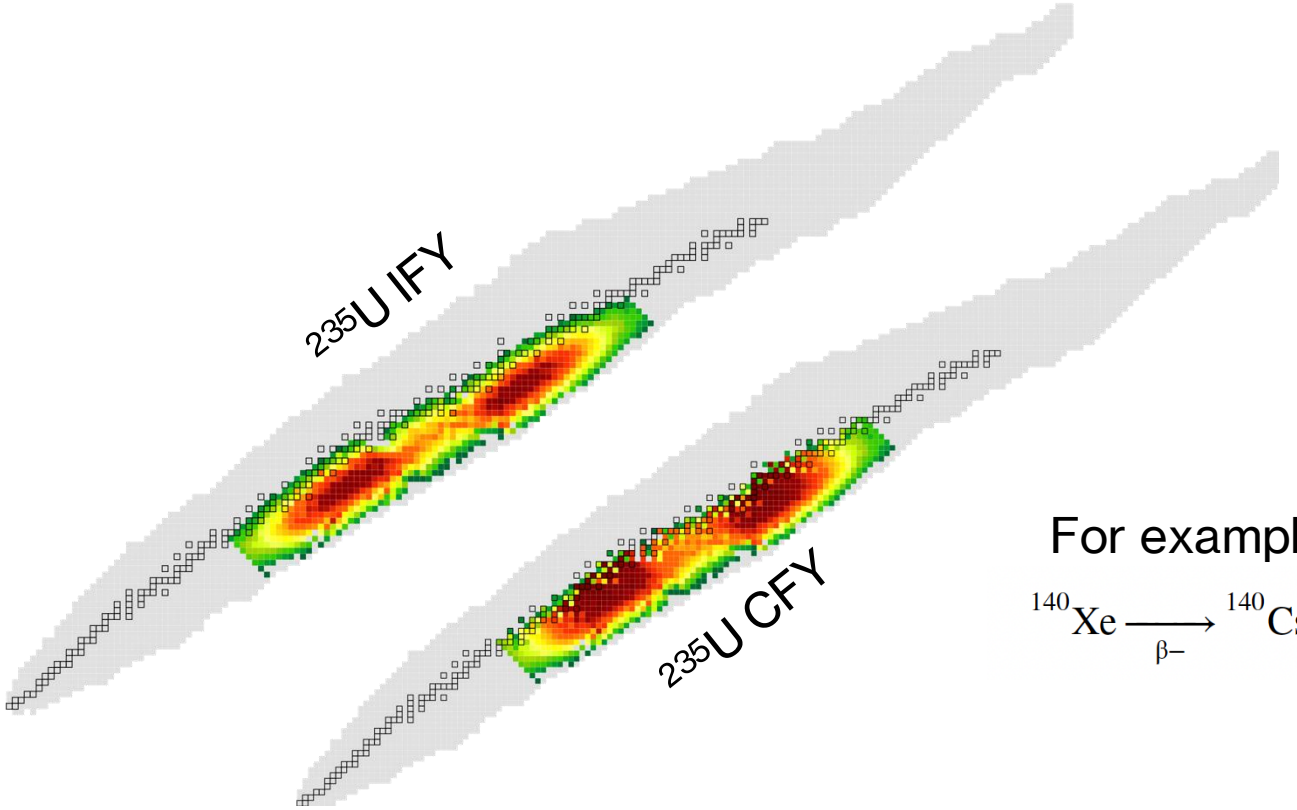


Spontaneous Fission of ^{252}Cf



What about yields?

- We typically refer to *cumulative yields* and *independent yields*
- *Independent Fission Yields (IFY)* are the yield of the nucleus as a primary fission product
- *Cumulative Fission Yields (CFY)* include all of the β-decays that occur after fission



For example,

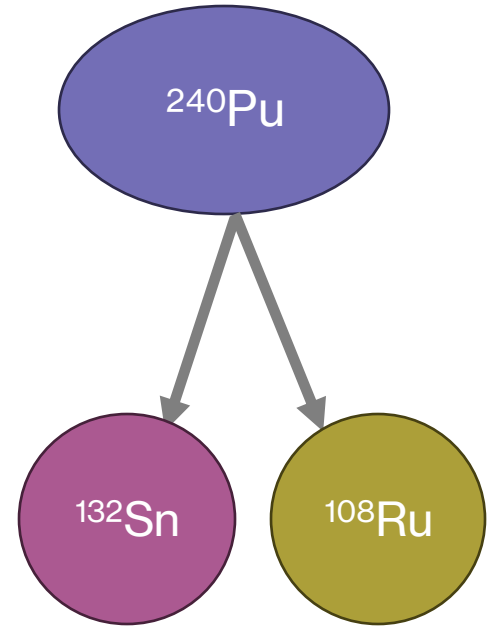
$$^{140}\mathrm{Xe} \xrightarrow[\beta-]{} {}^{140}\mathrm{Cs} \xrightarrow[\beta-]{} {}^{140}\mathrm{Ba} \xrightarrow[\beta-]{} {}^{140}\mathrm{La} \xrightarrow[\beta-]{} {}^{140}\mathrm{Ce}$$

Fission Fragments: Kinetic Energy

- The total kinetic energy of fission products is roughly related to the Coulomb repulsion between the two main fission fragments

$$TKE = \frac{Z_1 Z_2 \alpha \hbar c}{1.8\left(A_1^{1/3} + A_2^{1/3}\right)} MeV$$

Here $r_0 = 1.8\ fm$ because of the strong deformation present at scission
α is the fine structure constant

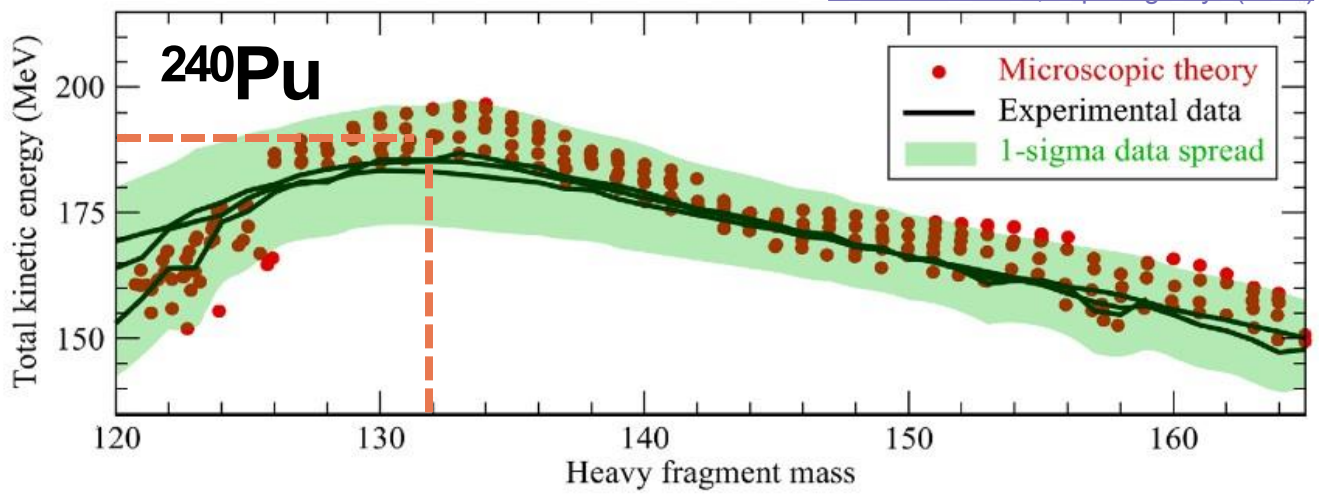


-For example, take ^{240}Pu $Z = 94, N = 146$
-If we use $Z_1 = 50, N_1 = 82$,

$$Z_2 = 94 - 50 = 44$$
$$N_2 = 146 - 82 = 64$$

$$TKE = \frac{(50)(44)\alpha\hbar c}{1.8(132^{1/3} + 108^{1/3})} = 178\ MeV$$

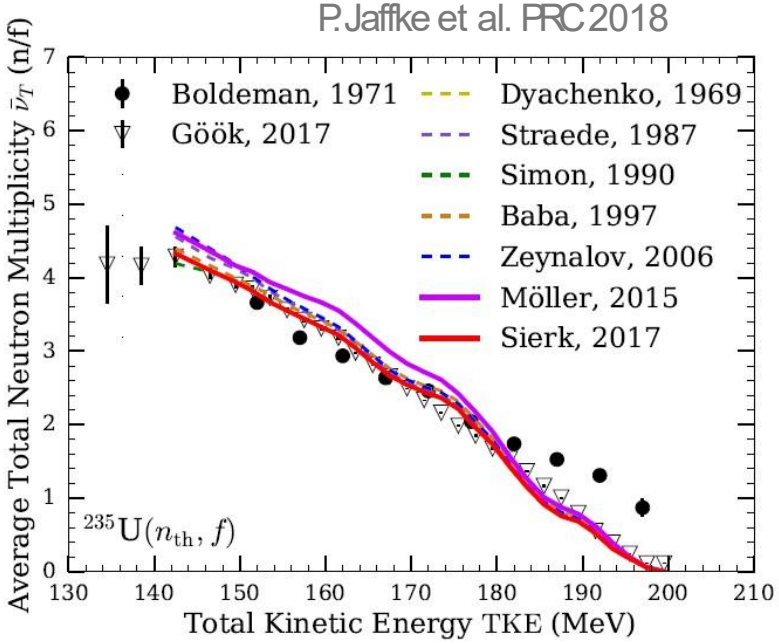
Schunk & Robledo, Rep. Prog. Phys. (2016)



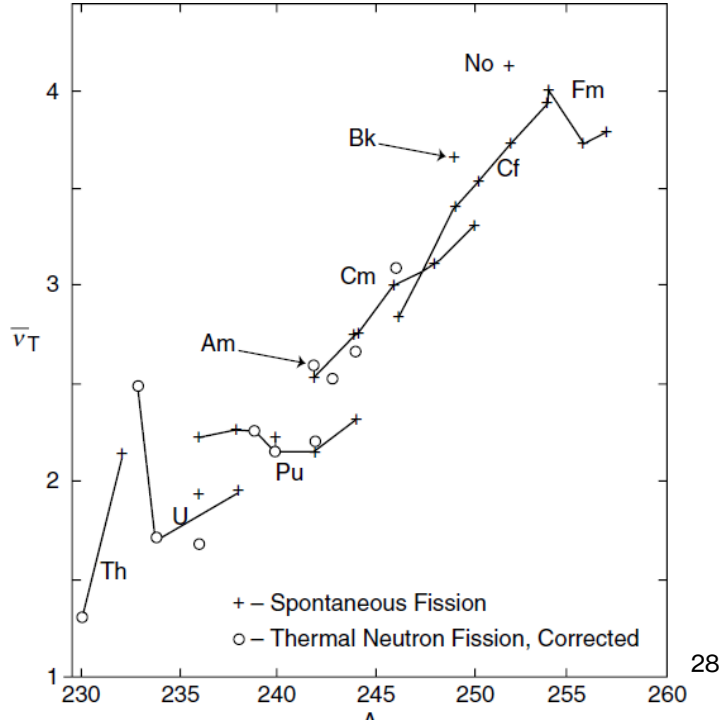
Neutron Emission

- The total excitation energy is obtained from energy conservation, $\boldsymbol{TXE} = \left[\boldsymbol{E_{cm}} + \boldsymbol{B_n} + \boldsymbol{Q_f}\right] - \boldsymbol{TKE}$ where the Q-value plus any reaction energy can pay for the fragment kinetic energy and to unbind a neutron
- Typically, $\boldsymbol{TXE}$ is on the order of 10s of MeV
- A rough estimate is that ~10 MeV of excitation energy is removed from γ emission and β-decay and the associated ν emission, and the rest is for neutrons
- It takes ~ 5MeV to create 1 neutron, so a few neutrons are emitted per fission
- Deviations from the linear trend with mass (TXE is higher with A) are mostly due to shell corrections

P.Jaffke et al. PRC 2018



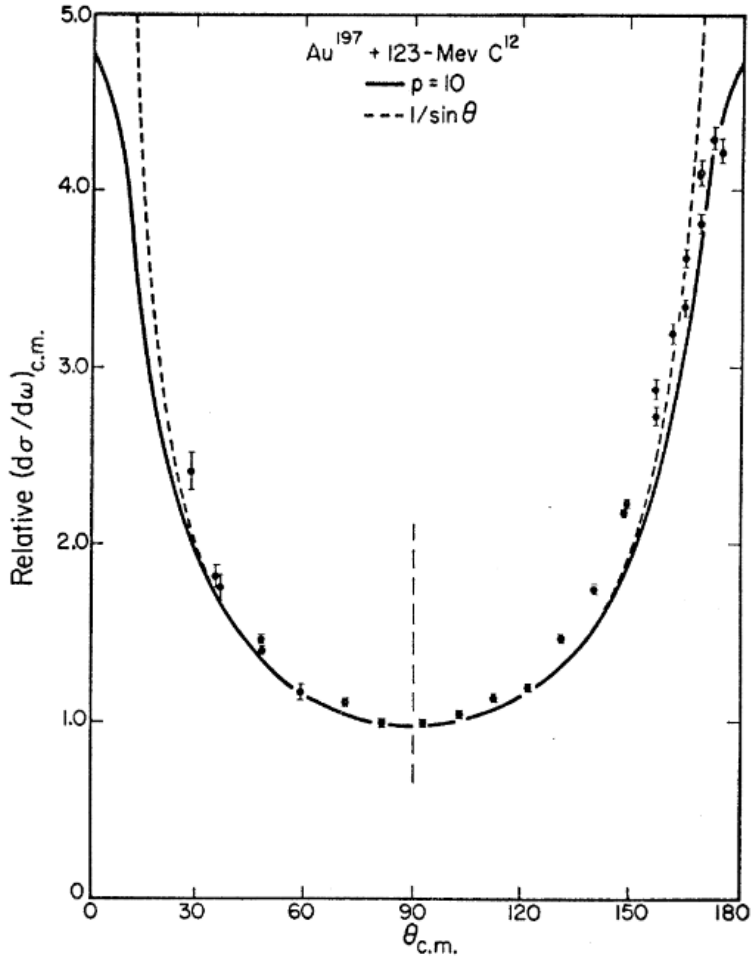
Loveland, Morrissey, & Seaborg, Modern Nuclear Chemistry (2006)



Fission fragment angular distributions

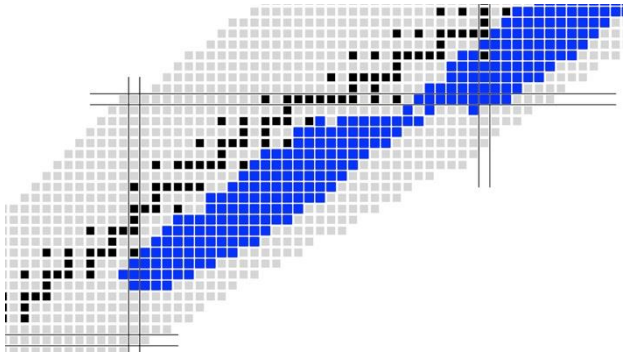
- Induced fission is slow enough that the nucleus reaches statistical equilibrium
- This results in an angular distribution of fission fragments that is symmetric about 90° in the center of mass, where 90° marks the plane perpendicular to the direction of motion of the fissioning system.
- This is because forward and backward angles maximize the amount of angular momentum the fragments carry away. Prior to scission, the fissioning nucleus will be highly excited relative to the absolute ground state, and higher excitations correspond on average to larger angular momenta.
- The proper formula for the angular distribution is complicated and related to the deformation of the fissioning nucleus. [See Loveland, Morrissey, & Seaborg]
- From momentum conservation, we know that the fragments themselves are emitted 180° apart (in the CM), which is handy for gating on fission events in reaction-induced fission studies

G. Gordon et al. Phys. Rev. 1960



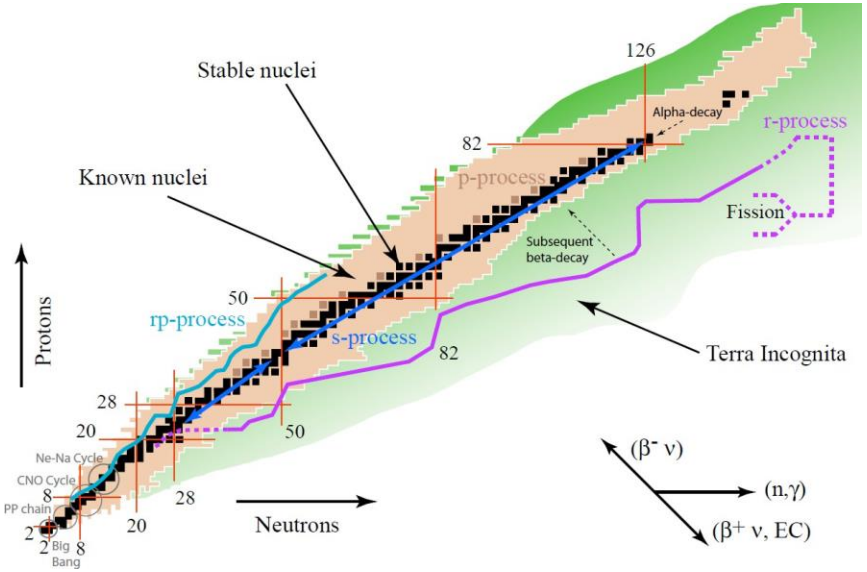
Why should you care?

- Nuclear Energy
- Nuclear Astrophysics
- Beam Production

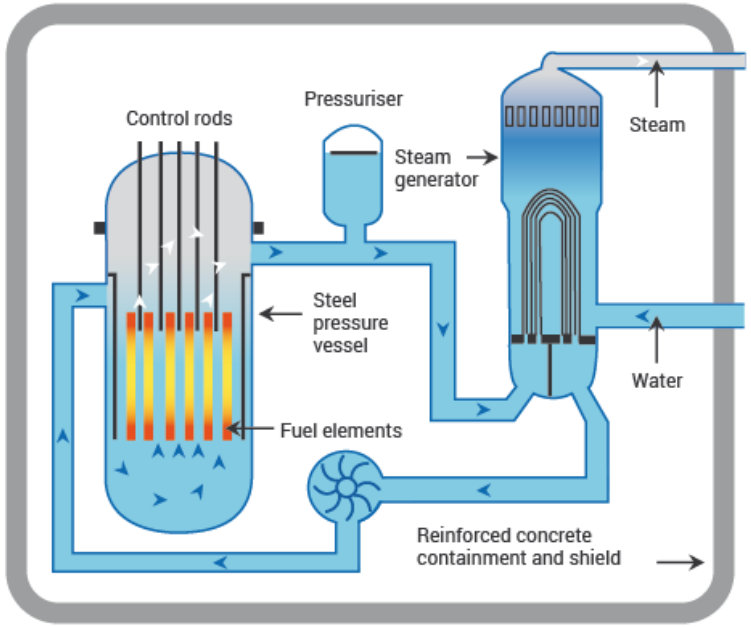


CARIBU/nuCARIBU at ANL

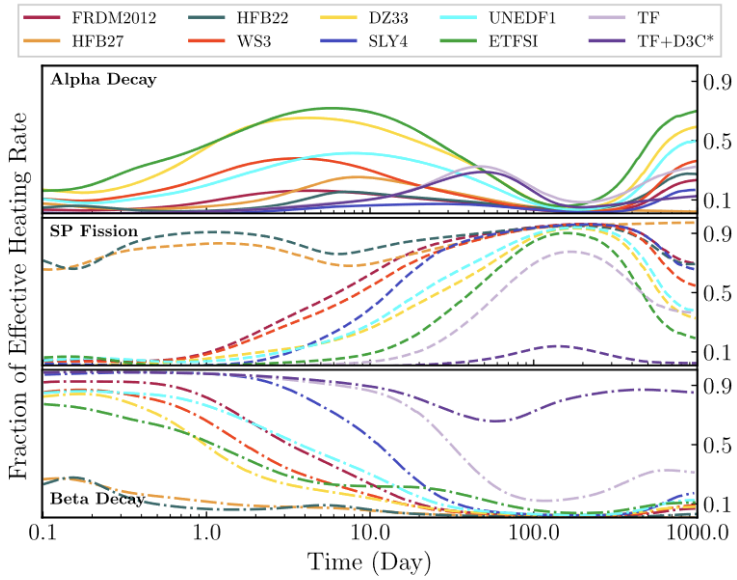




M. Smith et al. Nuclei in the Cosmos 2006 Conference Proceedings



World Nuclear Association



(b) $Y_e = 0.16$

Wrapping up!

- HW #3 posted and extended to Oct. 14th **at 11:59PM**
- No Office Hours this week due to Fall Break
- **Next class**: Nuclear Reactions
- Questions?